

Design Study of the Inner PSXM for Spiral Injection: Field Capability and Shield-Radius Constraint

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Abstract

The PSXM (Passive Shielding auXiliary Magnet) is a pulsed auxiliary magnet that steers the injected muon beam during three-dimensional spiral injection in the J-PARC muon $g-2$ /EDM experiment, producing a dipole or quadrupole field at its centre behind a conducting shield can. This note studies the inner PSXM, the passive counterpart of the ASSM [1], in two-dimensional magnetostatics, with an equal-weight least-squares solve for the coil currents.

At the 1000 A working-current cap adopted in this study the bare six-coil ring reaches 23.70 mT of central dipole field and 2.179 mT/mm of central gradient, and the quadrupole can be rolled to any angle about the beam axis. The shield counter-field reduces the central amplitude of multipole order m by the factor $1 - (R/R_s)^{2m}$, which inverts into an explicit design map for the shield radius, so the smallest ideal-sheet radius at which the adopted working benchmark $G^* = 1$ mT/mm remains deliverable is $R_s \simeq 26.2$ mm. The numerical solve reproduces that analytic ideal-shell value. The benchmark is a study assumption, not a beam-derived requirement, and the threshold should be re-evaluated when the beam-optics requirement and hardware current limit are finalised. The dipole radius is not fixed here: the provisional 1 mT benchmark is unbuildable at 22.990 mm, while the published field integral $BL = 1.0 \times 10^{-3}$ T m, a beam-deflection requirement, implies about 10 mT of central field under the 0.1 m effective length; settling it needs BL in a finite-length geometry.

The shielding model is an idealised two-dimensional reference problem, not a leakage prediction: its exterior values depend on the discretisation and sampling of the equivalent shield current and cannot verify the published 1 μ T requirement; quantitative shielding needs a finite-length, transient model.

Keywords: muon $g - 2$; electric dipole moment; spiral injection; steering magnet; passive shielding; two-dimensional magnetostatics; inverse problem

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1 Introduction

1.1 Role of the PSXM in the injection scheme

The J-PARC muon $g-2$ /EDM experiment stores a low-emittance 300 MeV/c muon beam in a compact, highly uniform solenoidal storage field. Because the storage magnet has no conventional injection straight, the beam is brought in along the axis and spirals down onto the storage plane over many turns, the three-dimensional spiral injection scheme [2, 3, 4, 5]. Along that path the beam passes a double bend achromat, a pair of rotated quadrupoles, the *outer* auxiliary magnet just before the iron yoke, and the *inner* auxiliary magnet inside it, before descending to the storage plane [7].

The inner auxiliary magnet is the subject of this note. It is a pulsed magnet on the injection path whose six main coils generate a dipole or quadrupole field at its centre, and its role is to steer and tune the injected beam, correcting the orbit and shaping the phase space before the beam reaches the kicker [7]. The passively shielded variant studied here, in which the coils are surrounded by an unpowered conducting can, is denoted PSXM, for Passive Shielding auXiliary Magnet.

Besides the fields above, other magnetic fields act on the same beam and are not part of the PSXM design problem: the fringe field of the storage solenoid, which sets the optics of the spiral; the pulsed radial kicker field near the storage plane, which removes the axial momentum accumulated during the descent; and the weak-focusing field, which provides the vertical confinement of the stored beam. In particular the PSXM does not stop the vertical motion at the mid-plane; the kicker does.

1.2 Active Shield Steering Magnet as the reference design

A steering magnet for this position has been designed before. Abe *et al.* [1] designed the Active Shield Steering Magnet (ASSM), in both an inner and an outer version, using a current-potential method adapted from MRI actively shielded gradient-coil design. Their paper states the role of the devices unambiguously: the injection volume carries *error fields* (from wall reinforcement iron, tolerances in the iron-yoke and magnet assembly, and so on), which disturb the injection orbit, and the steering magnets exist to compensate that disturbance. The outer magnet controls the beam direction relative to the tunnel axis; the inner magnet controls the axial velocity, that is the pitch angle, of the beam already on the spiral orbit. Stopping the axial motion at the mid-plane is done by the radial kicker, not by these magnets.

Because the steering magnets sit close to the iron yoke and the vacuum vessel, their own stray field would itself become an error field, which is why they need a shield at all and why the ASSM carries an actively driven superconducting shield coil. The PSXM is the passively shielded alternative, in which the induced currents of an unpowered can do the same job; the two share a purpose and a set of requirements while using different methods. Table 1 carries the ASSM specifications over to this study for quantitative comparison. Following the agreed plan of work, this note studies the inner magnet only and leaves the outer magnet to future work.

1.3 Design requirements

The design is governed by two requirements that act against each other. The six main coils, driven by short pulses of order 1 kA and $\lesssim 1 \mu\text{s}$ [7], must first produce a dipole or quadrupole field of a specified amplitude at the centre, uniform enough over the aperture that the whole bunch receives a consistent kick; the aperture over which that uniformity is required is a beam-based question, set by the transverse beam envelope, that remains open. At the same time that pulsed field must not reach the storage region, whose precision rests on the uniformity of the storage field. In the passive design considered here the second requirement is addressed by a

conducting shield can around the coils, which carries induced currents that cancel the exterior field. Making the central field larger and keeping the exterior field small are competing demands on the same geometry.

The exterior field is framed by two distances. The published requirement for the inner steering magnet is a leaked field below $1\ \mu\text{T}$ at 0.2 m [1]; I therefore evaluate the exterior field at 200 mm , and at a near-field diagnostic radius of about 419 mm , the closest approach of another beam segment on the injection arc to the magnet at $z \approx 1.1\text{ m}$ (fig. 1). The storage region is over 1 m away, so the 419 mm value serves only as a conservative diagnostic and does not enter the design conclusion. The present two-dimensional model does not quantitatively predict the physical leakage field and cannot verify the $1\ \mu\text{T}$ requirement.

Table 1: Published target specifications of the Active Shield Steering Magnets (ASSM) [1], carried over as the requirement set for this study. The PSXM is the passively shielded alternative to the inner magnet; the outer magnet is outside the scope of this note.

Parameter	Inner
First field integral BL (T m)	1.0×10^{-3}
Field direction (storage-magnet frame)	radial
Orbit control	pitch
Leaked field	$< 1.0\ \mu\text{T}$ at 0.2 m
Joule heat	$< 1.0\text{ W}$
Allowable diameter	100 mm
Allowable length	200 mm

Note: the paper’s text quotes the leaked field at 0.2 m as “less than a few μT ” (its Sec. III-C and Summary); the $1\ \mu\text{T}$ level follows the specifications in its Table I. Neither level is verified by the present model.

The size envelope in table 1 settles one question immediately: R_s is the radius of the ideal surface-current sheet, not a physical dimension of the can, and with the 5 mm baseline copper wall [6] even the larger illustrative dipole radius considered below fits comfortably inside the 94 mm envelope of the internal PSXM baseline [6], so the aperture is not binding. The binding constraint on the shield radius is therefore the available current rather than the aperture.

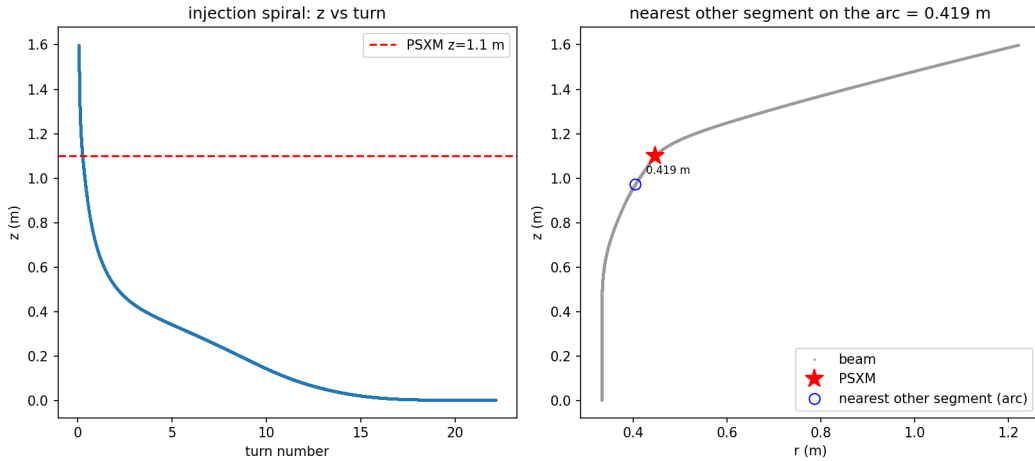


Figure 1: Injection spiral from the tracking data: vertical position versus turn number, and the closest approach of another beam segment to the injection magnet. The 419 mm point is the nearest other segment on the incoming arc at the magnet’s location; the storage region is over 1 m away.

The study addresses four questions in order: the dipole and quadrupole fields the six-coil geometry can produce at the centre and over what region; whether the quadrupole can be presented at any roll angle about the beam axis and at what cost; how the ideal shield representation behaves numerically and what its exterior residual depends on; and which shield radius follows from the field-delivery constraint.

2 The simulation model

2.1 Two-dimensional magnetostatics

Each current-carrying element is an infinite straight wire piercing the x - y plane at (x_0, y_0) with current I . Its field is the Biot–Savart result for a straight wire,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{2\pi\rho^2} (-\Delta y, \Delta x), \quad \Delta x = x - x_0, \quad \Delta y = y - y_0, \quad \rho^2 = \Delta x^2 + \Delta y^2, \quad (1)$$

and the total field is the linear superposition over all wires,

$$\mathbf{B} = K \mathbf{I}, \quad (2)$$

where K is a coefficient matrix in T/A and $\mathbf{I}$ the current vector. Column j of K is the field pattern produced at every sample point by 1 A in current degree of freedom j .

Within the present free-space, fixed-permeability model the field is exactly linear in the coil currents; the linearity is a property of that model, and ferromagnetic material in a real assembly would break it. The linearity is used repeatedly below: one solve per geometry plus a scale factor covers every operating current, and two solved current patterns can be combined to roll the quadrupole. Units are consistent throughout the codebase, with positions in mm, currents in A and fields in T; gradients are quoted in mT/mm, which is numerically identical to T/m. The field-line figures plot contours of the magnetic vector potential A_z , whose contour lines are exactly the field lines of eq. (1).

The infinite-wire idealisation fixes the domain of the model: it is accurate near the magnet mid-plane over a transverse scale of order the coil dimensions, whereas the exterior diagnostic radii defined above lie well outside it. The reported exterior-field values are therefore quantities of the two-dimensional model, and quantifying physical shielding requires a three-dimensional treatment that this study does not attempt.

The model is also a free-space one. The static 3 T storage field and the nearby ferromagnetic yoke and vacuum vessel are not part of it. The incremental pulsed field solves a linear problem in vacuum, and copper is non-magnetic, so the static background does not enter the solve, but magnetic boundaries near the magnet would modify the field and are a scope limit of this study.

2.2 Geometry of the six-coil ring

The PSXM places six physical coils evenly on a circle of radius $R = 22.5$ mm. Each coil is a racetrack winding whose two straight legs pierce the transverse plane at the same radius but at different angles, so a single physical coil appears in this cross-section as two wires carrying $+I_k$ and $-I_k$. Six physical coils therefore drive twelve current points, though not twelve independent currents. The grouping matrix M (`group_matrix`) encodes

$$\mathbf{I}_{\text{leg}} = M \mathbf{I}_{\text{coil}}, \quad M \in \mathbb{R}^{12 \times 6}, \quad (3)$$

so that the inverse problem is solved for the six real degrees of freedom instead of twelve fictitious ones. With a chord of 20 mm between a coil’s legs, the half-angle between them is 26.4° .

With shielding enabled, N_s current points are placed on a circle of radius R_s around the coils, representing the conducting shield can as a numerical discretisation of a continuous current

sheet. One consequence must be settled before any shield current is quoted: the current per element scales as $1/N_s$ and is therefore a property of the discretisation, whereas the linear current density

$$\mathcal{K}_s = \frac{N_s I_s^{\max}}{2\pi R_s} \quad (4)$$

is what the continuum limit fixes; the convergence scan below verifies that $\mathcal{K}_s$ is independent of N_s , so every shield current below is quoted as a sheet current in A/mm. Table 2 collects the parameters.

Table 2: Geometry and operating parameters.

Symbol	Meaning	Value	Source
R	coil-ring radius	22.5 mm	geometry
L_c	chord between a coil’s legs	20 mm	geometry
N	number of physical coils	6	baseline design
R_s	shield radius	27.5 mm (default)	design variable
N_s	shield discretisation	100–200	numerical
d_{out}	outer $B = 0$ -ring offset	5 mm (baseline)	numerical
d_{in}	inner $B = 0$ -ring offset	2.5 mm (baseline)	numerical
Δr	offsets in robustness scan	2, 5, 10, 20 mm; paired inner ring at $\Delta r/2$	numerical
$I_{\max}$	coil current cap	1000 A	working assumption
–	pulse width	$\lesssim 1 \mu\text{s}$	working assumption
B_0^*	dipole working benchmark	1 mT	adopted benchmark
G^*	quadrupole working benchmark	1 mT/mm	adopted benchmark
–	near diagnostic radius	200 mm	published distance
–	far diagnostic radius	419 mm	nearest arc segment

2.3 The least-squares inverse problem

The design question is never what field a given set of currents makes, but which currents make a prescribed field. The solver (`current_solver.py`) supports the general weighted least-squares formulation below; the calculations in this report use its equal-weight case. Sample points carry a prescribed target field (B_x^*, B_y^*) , and the solver minimises

$$\mathbf{I} = \arg \min_{\mathbf{I}} \sum_j w_j |\mathbf{B}_j - \mathbf{B}_j^*|^2, \quad \mathbf{B} = K\mathbf{M}\mathbf{I}. \quad (5)$$

The unknowns of eq. (5) live where the conductors are, while the constraints live where the field is specified, and nothing forces the two sets to coincide or even to have the same size. That separation is what makes the formulation a design tool rather than a field calculator, and it is used deliberately below: there are always more field constraints than current unknowns, so eq. (5) is a genuine least-squares problem with a small but non-zero residual rather than a square system with an exact solution. A six-coil magnet cannot reproduce an ideal multipole exactly, and that residual is the honest measure of how close it gets.

The implementation supports a non-negative weight w_j for each sample point: both component rows for point j are multiplied by $\sqrt{w_j}$ before the matrix solve. No custom weights are passed in the calculations reported here, so $w_j = 1$ for every point and the reported results are the ordinary equal-weight least-squares case. The central-field coil solve and the shield-response solve are performed separately; the 1 mm central ring is therefore never mixed in one objective with the exterior $B = 0$ rings. The solver returns the raw, unnormalised solution, and rescaling to the adopted current cap is a separate explicit step, so that a single solved current vector

stays consistent across the predicted field, plotted field and reported exterior field. Figure 2 shows the field lines of one solved configuration.

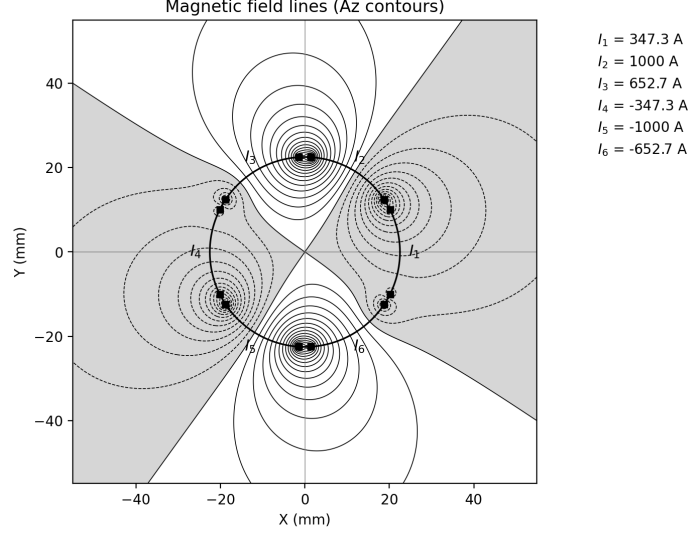


Figure 2: Field lines (A_z contours) of a solved six-coil configuration, with the coil-ring boundary and the $I_1 \dots I_6$ labels. Solving for the six physical currents rather than twelve per-leg currents is what makes the solution physically realizable.

2.4 Ideal shield response in the two-dimensional model

Requiring the total field to vanish on sampling rings offset outward from the shield and solving eq. (5) for the shield block alone gives a fixed linear map between the coil currents and the shield currents,

$$K_s \mathbf{I}_s \simeq -K_6 \mathbf{I}_c \implies \boxed{\mathbf{I}_s = S \mathbf{I}_c}, \quad S = -K_s^+ K_6, \quad (6)$$

where K_6 and K_s are the coefficient blocks of the coil and shield degrees of freedom and $+$ denotes the least-squares inverse; since S depends only on geometry, the ideal shield response is fixed once the six coil currents are. This is an ideal equivalent surface current, not a prediction of the physical eddy-current dynamics. In the baseline configuration Δr is the radial offset of the outer zero-field ring: $\Delta r = 5 \text{ mm}$, while the paired inner ring is at $\Delta r/2 = 2.5 \text{ mm}$. Thus the two radii are $R_s + 2.5 \text{ mm}$ and $R_s + 5 \text{ mm}$. Nothing is imposed on the shield surface itself; the sample points are placed in free space to avoid the point-current singularities of the discrete representation. In the continuous perfect-flux-exclusion limit with zero net enclosed current the exterior incremental field vanishes exactly, so the finite- N_s exterior values reported below are numerical residuals of that limit. The physical limits of the ideal formulation are set out with those results.

2.5 Analytic cross-check of the ideal-shell result

The multipole argument above has a closed-form consequence that the numerical solve reproduces almost exactly: the central amplitude of an azimuthal harmonic of order m is reduced by the factor

$$\boxed{F_m(R_s) = F_{m,\text{bare}} \left[1 - \left(\frac{R}{R_s} \right)^{2m} \right]}, \quad (7)$$

and inverting it gives the threshold radius of eq. (15) directly,

$$R_s = R \left(1 - \frac{F_{\text{target}}}{F_{\text{bare}}} \right)^{-1/(2m)}, \quad (8)$$

for any target amplitude $F_{\text{target}} < F_{\text{bare}}$. With $R = 22.5$ mm and the bare amplitudes reported below, eq. (7) reproduces 7.836 mT and 1.203 mT/mm at $R_s = 27.5$ mm, and eq. (8) the numerical thresholds 22.990 and 26.233 mm, to the three decimal places quoted; the same-layout analytic and numerical thresholds coincide to better than 10^{-4} mm. Because the centre field is linear in the currents, eq. (8) becomes the explicit design map

$$R_{s,\min}(G^*, I_{\max}) = 22.5 \text{ mm} \left[1 - \frac{G^*}{2.179 \text{ mT/mm} (I_{\max}/1000 \text{ A})} \right]^{-1/4}, \quad (9)$$

for the quadrupole; the threshold 26.233 mm quoted below is an operating point of this map at the adopted benchmark and current cap rather than an independent design conclusion.

3 Dipole and quadrupole field capability

This section examines the strength, spatial extent and direction of the fields the magnet can produce. All numbers here are measured in the same way. I sample B on a small ring about the centre and fit

$$B_x = B_{0x} + G_s x + G_n y, \quad B_y = B_{0y} + G_n x - G_s y, \quad (10)$$

which gives the dipole vector (B_{0x}, B_{0y}) , the normal and skew gradients G_n and G_s . For N equally spaced samples on a ring of radius r , let $\delta B_{x,i}$ and $\delta B_{y,i}$ be the measured field minus this fitted dipole-plus-linear-quadrupole field. The absolute RMS higher-order residual is

$$\sigma_{\text{HO}}(r) = \left[\frac{1}{2N} \sum_{i=1}^N (\delta B_{x,i}^2 + \delta B_{y,i}^2) \right]^{1/2}, \quad (11)$$

which has units of tesla. All residual values quoted and plotted in this report are instead the relative RMS quantity

$$\epsilon_{\text{HO}}(r) = \frac{\sigma_{\text{HO}}(r)}{\left[\frac{1}{2N} \sum_{i=1}^N (B_{x,i}^2 + B_{y,i}^2) \right]^{1/2}}, \quad (12)$$

which is dimensionless. It is the field content not represented by the constant dipole and linear normal/skew quadrupole terms in eq. (10). Putting every achieved-field number on this common footing lets the dipole and rotation studies be compared directly.

3.1 Dipole mode

The dipole is the primary mode, since steering requires a field that is the same across the bunch. Solving eq. (5) for the target $B_x = B_0$, $B_y = 0$ on a 1 mm ring and renormalising to the 1000 A operating point gives a central field of 23.70 mT on the bare coil ring, about a factor of 24 above the 1 mT working benchmark, so the coil ring itself is not what limits the design. Any direction in the transverse plane is reachable, and the direction of (B_{0x}, B_{0y}) follows the target directly, as the direction scan of fig. 5 shows.

The good-field region is set by the coil radius. The solved bare-ring dipole deviates from its central value by 6.7 % at $r = 10$ mm, and the deviation grows steeply beyond that as the field point approaches the winding at 22.5 mm. The 1 mm fitting ring is a numerical region rather than a beam-derived good-field aperture; the field quality is reported as a function of radius below, and a beam-based aperture can be fixed once the beam envelope is known.

3.2 Quadrupole mode

For the quadrupole target $B_x = Gy$, $B_y = Gx$ the same procedure gives 2.179 mT/mm at 1000 A on the bare ring, again above the 1 mT/mm working benchmark, and the dimensionless relative residual ϵ_{HO} of eq. (12) is 2.4×10^{-3} on the 1 mm ring (the shielded angular scan below has a slightly larger maximum). The field quality deteriorates as the sampling radius is increased, which is expected because the observation points approach the conductors at $R = 22.5$ mm, where higher-order components become increasingly important. Over the sampled range the residual grows by about a factor of four when the sampling radius doubles, as expected when the leading contaminating term lies two multipole orders above the quadrupole: it is 2.39×10^{-3} at $r = 1$ mm, 9.56×10^{-3} at $r = 2$ mm, 5.96×10^{-2} at $r = 5$ mm and 2.32×10^{-1} at $r = 10$ mm. The useful linear range of the quadrupole is therefore determined mainly by the coil geometry rather than by the solver. The present study characterises the local multipole quality as a function of radius; translating this into a beam-based good-field aperture requires the transverse beam envelope and the orbit-correction tolerance, which are not part of this note.

3.3 Maximum field capability

The least-squares designs above constrain field quality by matching an ideal dipole or quadrupole at every point of the 1 mm sampling ring. To compute a hardware ceiling, the off-centre matching equations are not solved and no bound is imposed on ϵ_{HO} or dipole uniformity. Instead, for each field direction I maximise only the corresponding fitted central dipole component or fitted linear-gradient component, subject to the six independent bounds $|I_k| \leq I_{\text{max}}$. This box-constrained calculation gives 31.60 mT and 2.517 mT/mm in the most favourable direction; they are upper bounds, not field-quality-controlled operating designs. With the shield can in place the achievable values fall to 7.836 mT and 1.203 mT/mm at the default $R_s = 27.5$ mm (table 3); the shield, and not the coil current cap, therefore sets the operating point.

3.4 Field quality versus radius and the application range

The field quality above is quoted on the 1 mm ring used by the inverse solve. To show how it changes across the aperture, I repeat the fit of eq. (10) on rings of increasing radius. Figure 3 plots the actual relative residual $\epsilon_{\text{HO}}(r)$, not a ratio to its value at 1 mm. On the bare ring the quadrupole residual rises from 2.39×10^{-3} at 1 mm to 9.56×10^{-3} at 2 mm, 5.96×10^{-2} at 5 mm and 2.32×10^{-1} at 10 mm; the shielded curve follows the same trend. The bare dipole starts lower, at 6.53×10^{-6} on the 1 mm ring, and reaches 6.52×10^{-2} at 10 mm.

The application range therefore follows from the same scan: the useful linear region is determined mainly by the coil geometry rather than by the solver, and translating the quality-versus-radius curve into a beam-based good-field aperture requires the transverse beam envelope, which remains an open input.

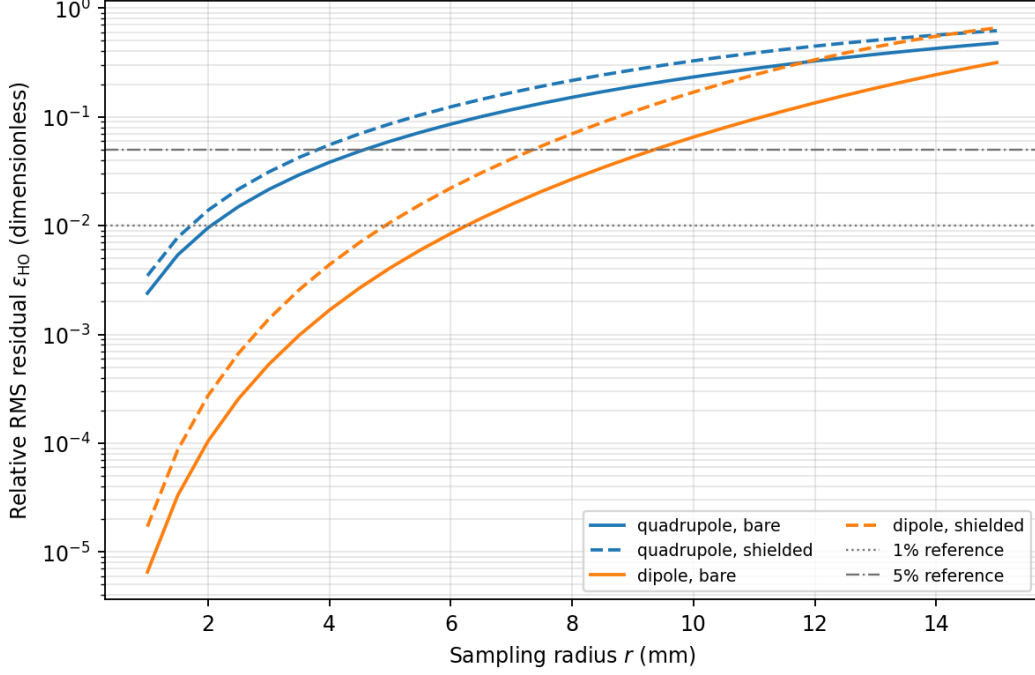


Figure 3: Actual dimensionless relative RMS higher-order residual $\epsilon_{\text{HO}}(r)$ from eq. (12) versus sampling radius, for the dipole and quadrupole with and without the default shield. The horizontal 1 % and 5 % lines are visual reference levels, not beam-derived requirements.

4 Rotational quadrupole

The magnet must be able to present its quadrupole at an arbitrary roll angle φ about the beam axis. This section establishes how the currents depend on φ and what the roll costs in amplitude and field quality.

4.1 Linear construction of a rotated quadrupole

A quadrupole rotated mechanically by φ is the target

$$B_x = G(y \cos 2\varphi - x \sin 2\varphi), \quad B_y = G(x \cos 2\varphi + y \sin 2\varphi), \quad (13)$$

which is $\cos 2\varphi$ times the normal-quadrupole target plus $\sin 2\varphi$ times the skew one; the angle appears doubled because a quadrupole pattern maps onto itself under a 180° mechanical rotation. Since the geometry does not move and the least-squares solution of eq. (5) depends linearly on the target, the solved currents should combine in the same way,

$$\mathbf{I}(\varphi) = \cos 2\varphi \mathbf{I}_{\text{normal}} + \sin 2\varphi \mathbf{I}_{\text{skew}}. \quad (14)$$

To test this I re-solved eq. (5) from scratch at every angle and compared against eq. (14); the discrepancy is 4.2×10^{-11} relative to the current scale, which is solver round-off. Operationally this is the useful result, since the magnet then needs two stored current patterns and one control parameter instead of a solve per angle, and the roll can be swept continuously. The same argument applies to the shield: the response matrix S of eq. (6) is fixed by geometry, so $\mathbf{I}_s(\varphi) = S \mathbf{I}(\varphi)$ inherits the same structure and the shield does not need to know that the field has been rotated.

4.2 Angular dependence of the achievable gradient

What is not rotation-invariant is the amplitude. Panel (b) of fig. 5 shows the achievable gradient at the 1000 A operating point as φ is scanned. It varies between 2.179 mT/mm at $\varphi = 0^\circ$ modulo 30° and 1.887 mT/mm at $\varphi = 15^\circ$ modulo 30° , a spread of 15.5 % repeating every 30° . The origin is geometric: six coils spaced 60° apart do not form a continuous current sheet, so some field orientations are cheaper in current than others, and the quadrupole's doubled angle halves the 60° lattice period to 30° in φ . For the present six-coil geometry the ripple is intrinsic; changing the number or angular arrangement of the coils would modify it. With the default shield can at $R_s = 27.5$ mm the same scan runs from 1.203 mT/mm down to 1.042 mT/mm: the shield counter-field reduces the central gradient by about 45 % while leaving the shape of the ripple unchanged, as linearity and the rotational symmetry of the circular shield require. The peak shield sheet current over the scan reaches $\mathcal{K}_s = 100$ A/mm (eq. (4)), which at $N_s = 200$ is 86 A per element. The $\cos 2\varphi / \sin 2\varphi$ construction is exact for the main-coil linear system, and the ideal circular shield preserves it; asymmetries of a physical can (a longitudinal slit, feedthroughs, asymmetric end geometry) would break the rotational symmetry of the response and could introduce an additional angle dependence.

The fitted quadrupole component rotates exactly by construction, since the least-squares solution is linear in the target; over the full scan the recovered angle tracks the commanded angle to within 4.3×10^{-14} degrees and the higher-order content stays below 3.5×10^{-3} in relative terms.

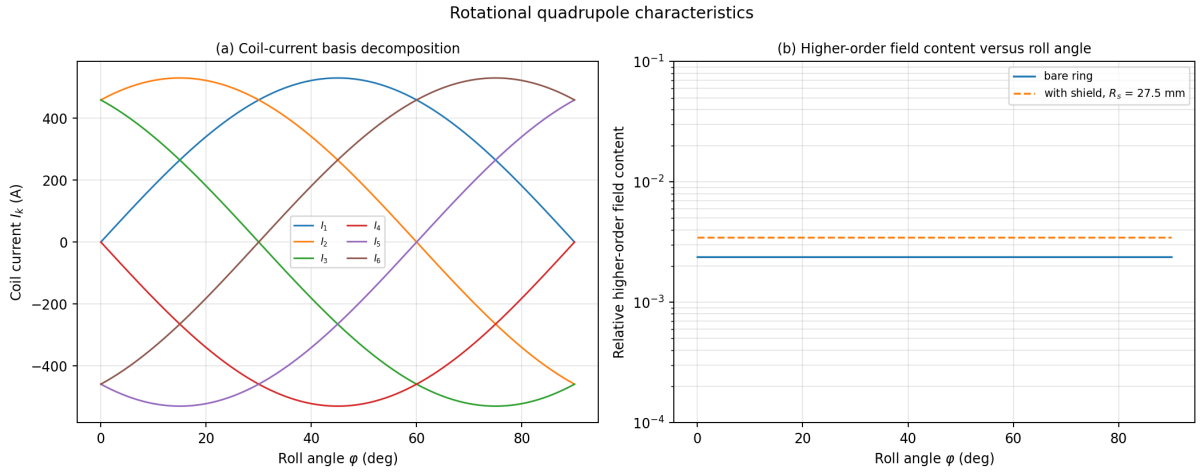


Figure 4: Rotational quadrupole characteristics. (a) The six solved coil currents versus commanded roll angle, where the $\cos 2\varphi$, $\sin 2\varphi$ law of eq. (14) holds exactly. (b) The relative higher-order content of the fit eq. (10), with and without the shield, held flat across the scan. The achievable gradient versus φ is shown in fig. 5(b).

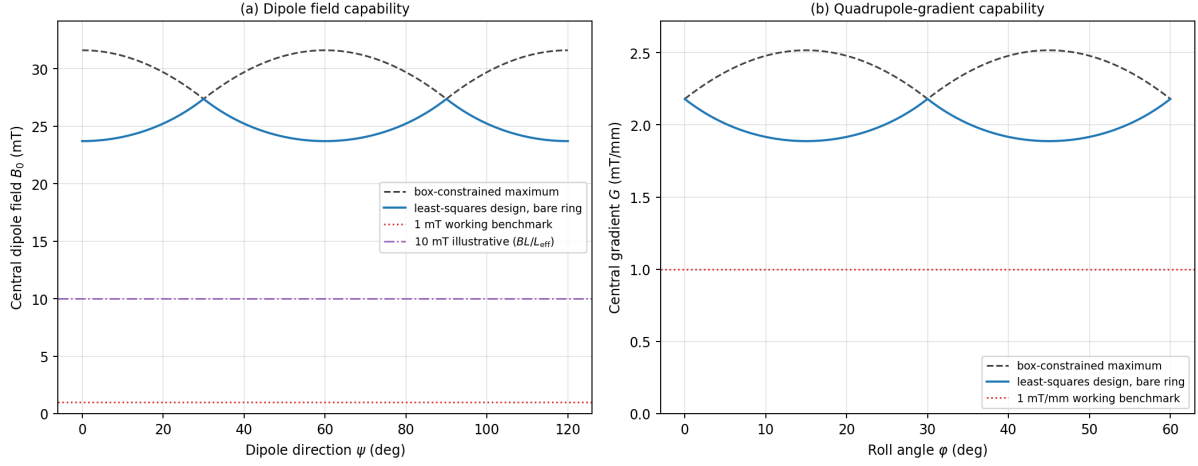


Figure 5: Central field capability at the 1000 A operating point: box-constrained maximum and least-squares design versus field direction for (a) the dipole and (b) the quadrupole, with the 1 mT/1 mT/mm working benchmarks. The design curve and the box-constrained maximum ripple in antiphase and coincide where the latter is smallest; at the reference orientation the design sits at 75 % and 100 % of the maximum, and comparing across orientations, 87 %, measures the ripple instead.

5 Scope of the ideal shield model

The purpose of this section is to delimit what the two-dimensional model can and cannot say about the exterior field; it does not predict the physical leakage field of the PSXM, and none of the exterior values below enters the shield-radius choice as a requirement. The exterior field is quantified as the ring-averaged $|\Delta B|$ at a given radius, with and without the shield for the same coil currents; the ring average is a scalar numerical diagnostic of the ideal-shell solution, whereas a physical requirement would be written against the field maximum on the relevant trajectory.

The ideal reference problem behind the numerical shield is the infinitely long, perfectly flux-excluding cylindrical shell. With zero net enclosed current the exterior incremental field vanishes exactly in the continuum limit, so any non-zero exterior field obtained numerically is a discretisation residual of the finite-current representation rather than the leakage field of a physical copper shield. Figure 6 shows the ring-averaged exterior field with and without the shield for both modes, in μT , with the published $1 \mu\text{T}$ level drawn for scale; the shielded curves are residuals relative to the ideal continuous-sheet solution. The published $1 \mu\text{T}$ requirement is accordingly not verified here, and neither diagnostic radius lies inside the validity domain of the model.

A convergence scan over the shield discretisation N_s , the azimuthal sampling density and the radial placement of the zero-field rings separates the converged design quantities from the numerical residuals. The delivered central gradient settles on 1.203 mT/mm and the required shield sheet current of eq. (4) settles on 100 A/mm at both $N_s = 100$ and $N_s = 200$, whereas the exterior residual at 419 mm falls by ten orders of magnitude as the current sheet is refined and never converges to a finite, discretisation-independent value. The first two are therefore stable quantities of the ideal-sheet model, and the third is a discretisation residual rather than a physical leakage prediction. Adding the shield also changes what the magnet delivers at the centre, and the cost is larger for the dipole than for the quadrupole: Table 3 collects, for both modes at the 1000 A operating point, the central field on the bare ring and behind the default shield at $R_s = 27.5$ mm, and the reduction follows exactly from eq. (7).

Since the ideal two-dimensional shield cancels the exterior incremental field exactly, this

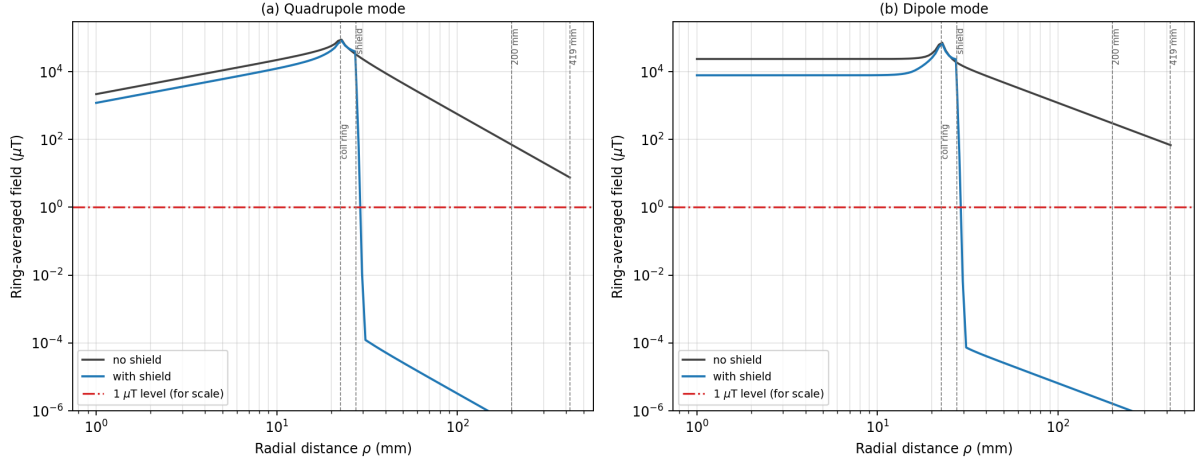


Figure 6: Ring-averaged exterior field versus radius at the 1000 A operating point, with and without the shield, for (a) the quadrupole and (b) the dipole design. Vertical lines mark the coil ring, the shield and the two diagnostic radii; the shielded curves are discretisation residuals relative to the ideal continuous-sheet solution.

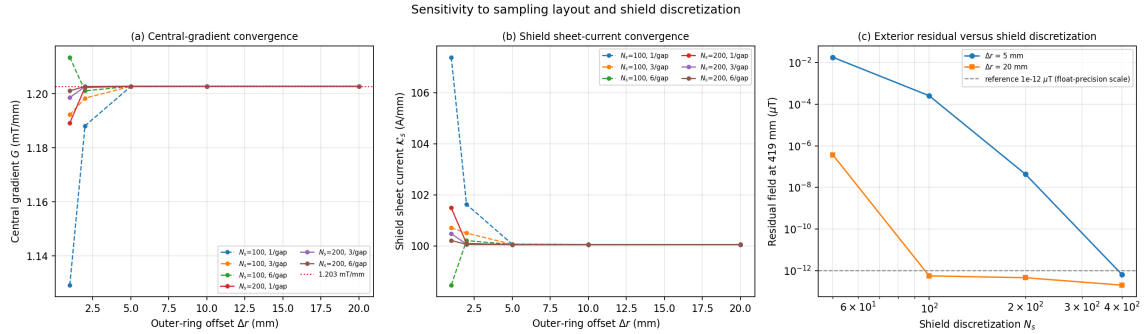


Figure 7: Discretisation scan at $R_s = 27.5$ mm: the delivered central gradient (a) and the required shield sheet current (b) are flat in the discretisation N_s and the ring offset Δr once $\Delta r \gtrsim 5$ mm, whereas the exterior residual at 419 mm (c) falls by orders of magnitude as the current sheet is refined and never converges.

Table 3: Central-field change when the shield is added, at the 1000 A operating point. “Bare” is the six-coil ring without the shield; “shielded” is the same ring behind the default ideal shield at $R_s = 27.5$ mm. The ratio and the percentage reduction measure the price of passive shielding at the centre.

Mode	Bare	Shielded	Reduction
Quadrupole gradient G (mT/mm)	2.179	1.203	$\times 1.81$ (44.8%)
Dipole central field B_0 (mT)	23.70	7.836	$\times 3.03$ (66.9%)

model carries no information about the performance of a physical shield. The mechanisms that limit a real passive shield lie outside it: flux exclusion is only approximate at finite conductivity and finite pulse length, the can has a finite axial length so the field leaks around its ends, and any longitudinal slit modifies or interrupts the closed current paths required to realise the ideal axial sheet-current distribution in a finite conducting can. Each of these can only reduce the suppression relative to the ideal limit.

The literal form of the ideal-conductor condition is also worth testing as an alternative formulation. With the present point-wire discretisation and collocation scheme, imposing eq. (19) directly at points on the shield ring is badly conditioned, because the normal component at a point on the ring is dominated by the near field of the two adjacent wires and not by the smooth part of the field. The resulting solve leaves a fraction 0.40 of the shell's quadrupole mode of A_z , against 3×10^{-4} for the free-space rings. Measured on the physical condition it is meant to enforce, the direct form is some three orders of magnitude worse than the indirect one in this implementation, and the free-space rings are retained for that reason.

6 Shield-radius constraint from field delivery

6.1 Field-delivery constraint

A larger shield can might be expected to shield better. The present model addresses only one half of that question. A shield close to the coil ring couples strongly to the coils and produces a stronger counter-field in the central region, so a larger main-coil current is required to maintain the same target field, whereas a shield far from the coil ring barely perturbs the centre and the required current is low. This half of the model is converged, but it provides a lower feasibility bound on the shield radius rather than a unique optimum: larger radii remain feasible and give greater field headroom or a lower required main-coil current, and selecting among them requires a physical shielding and mechanical model. Whether the exterior field is better or worse suppressed at one radius than another is not accessible here, since in the ideal two-dimensional limit the exterior incremental field vanishes at any radius; for a real shield that ranking will be set by finite conductivity, finite length and the can geometry. The scan below therefore constrains field delivery under the current cap and treats the exterior residual as a diagnostic.

6.2 Threshold definition

This study adopts a working cap of 1000 A on the coil current, reflecting the PSXM working assumption of a peak current of order 1 kA [7]; it is not a verified hardware limit. For each candidate shield radius R_s I solve eq. (5) for the target central field, read off the required current $I_{\text{req}}(R_s) = \max_k |I_k|$, and record the exterior residual. The design satisfies the current cap only where $I_{\text{req}} \leq 1000$ A, so the smallest usable radius is

$$\boxed{R_s^{\min} = \min \{ R_s : I_{\text{req}}(R_s) \leq I_{\max} \} ,} \quad (15)$$

a field-delivery threshold within the present two-dimensional inverse-design model and not a verified hardware optimum. Because the system is linear, one solve per radius plus a scale factor covers the whole scan: every number below comes from a single scan at the 1000 A operating point, and the required current at any other target follows by a common scaling of the currents.

Since the central field increases monotonically with R_s , equivalently I_{req} falls monotonically, the feasible region has a well-defined lower boundary. A radius smaller than $R_s^{\min}$ cannot reach the target; larger radii remain feasible and provide increasing field headroom at the cost of aperture. Nothing on the shielding side enters eq. (15); the exterior residual is a property of the discretisation.

6.3 Results

I scanned the shield radius in 0.4 mm steps from 22.8 mm to 60 mm while keeping the peak main-coil current at 1000 A. Figure 8 shows both modes. As R_s increases the delivered field rises rapidly at first and then flattens toward the bare-ring limit. The gradient crosses the adopted 1 mT/mm working benchmark at $R_s \simeq 26.2$ mm (26.233 mm in the numerical model), which is the threshold of eq. (15) for the quadrupole. The numerical threshold agrees with the analytic ideal-shell value of eq. (8) to better than 10^{-4} mm.

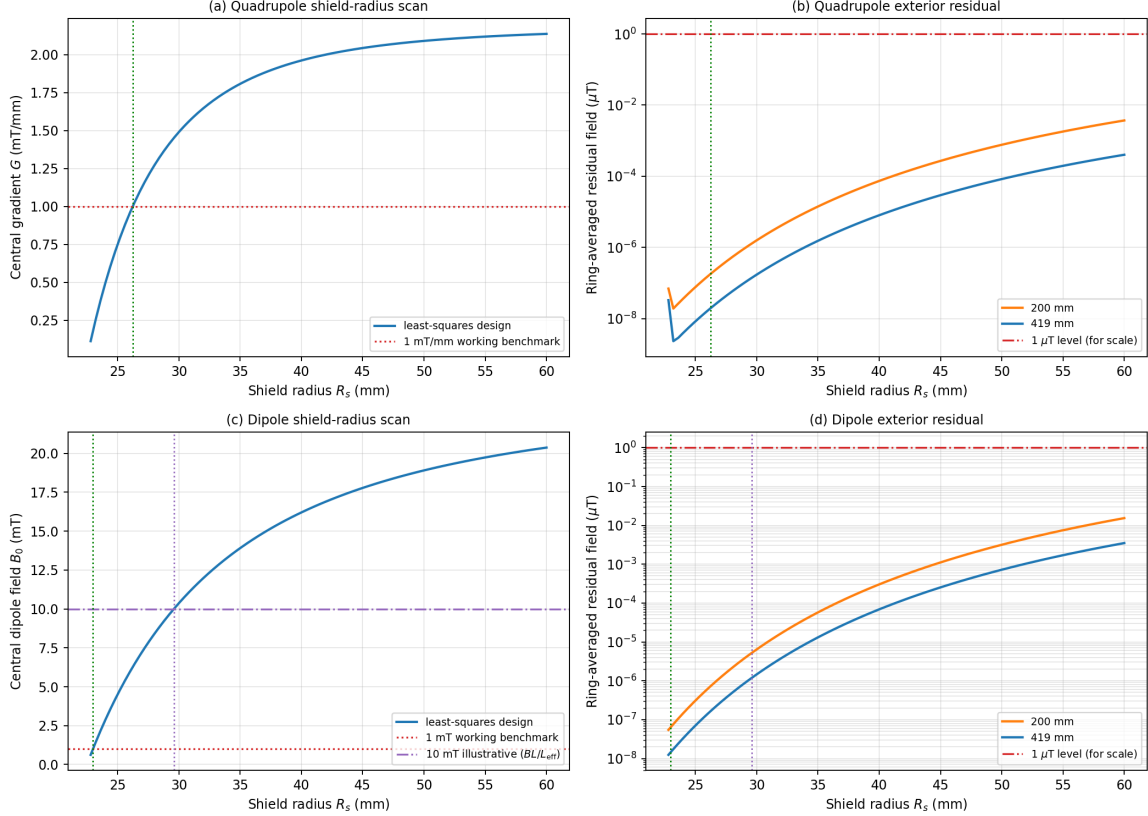


Figure 8: Shield-radius scan at the 1000 A operating point. (a) and (c) show the central field delivered in quadrupole and dipole mode against the corresponding criteria, with the crossings the thresholds of eq. (15); (b) and (d) show the ring-averaged exterior residual at the two diagnostic radii, a model quantity (section 5) and not a predicted field level.

6.4 Dipole criterion based on the field integral

For the quadrupole the criterion is unambiguous, and eq. (15) gives

$$R_s^{\text{quad}} \approx 26.2 \text{ mm}, \quad (16)$$

numerically 26.233 mm in the present model, the analytic ideal-shell value of eq. (8), which leaves a radial separation of about 3.7 mm between the coil-current locus and the ideal shield surface. For the dipole the answer depends on how the requirement is written, and the published material does not define a central-field target by itself.

Taking the provisional 1 mT central-field working benchmark at face value gives the radius $R_s^{\text{dip}} = 22.990$ mm. This leaves only 0.49 mm between the modelled coil-current locus and the ideal shield surface at $R = 22.5$ mm, which is unlikely to accommodate the finite conductor cross-section, insulation and mechanical support; read on its own, the result forces an engineering compromise at $R_s \in [23.5 \text{ mm}, 24 \text{ mm}]$, delivering 1.97 mT to about 2.9 mT.

The requirement actually published for the inner steering magnet is a first field integral, $BL = 1.0 \times 10^{-3} \text{ T m}$ (table 1), which sets the beam deflection directly and is therefore the physically relevant dipole criterion. It is not a central field: converting it requires the effective magnetic length, which the published material quotes as 0.1 m [1, 5]. The physical length of the three-dimensional device, 164 mm [1], is not an effective magnetic length and must not be used in this conversion. Under the 0.1 m reading, eq. (15) gives

$$R_s^{\text{dip}, BL} \approx 29.6 \text{ mm}, \quad (17)$$

about 3.4 mm beyond the quadrupole threshold. That number is an illustrative sensitivity estimate rather than a design conclusion: it assumes the PSXM inherits the inner magnet’s role and field integral, and it still replaces the field integral by a central-field proxy. The dipole radius is therefore left open here; it should be settled by evaluating $BL = \int B_y dl$ in the finite-length geometry, or by confirming the PSXM effective length, both of which are three-dimensional work.

6.5 Recommendation

Under the adopted working benchmark $G^* = 1 \text{ mT/mm}$ and the 1000 A working-current cap, the ideal-sheet field-delivery threshold is $R_s \approx 26.2 \text{ mm}$ (eq. (9)), and the default geometry of the simulation, $R_s = 27.5 \text{ mm}$, is retained as the reference geometry. The threshold is a conditional constraint of the present model rather than a final design radius: it should be re-evaluated when the beam-optics requirement and the hardware current limit are finalised, and the physical optimum will also depend on shielding, thermal and mechanical behaviour that this model does not address. With the 5 mm baseline copper wall [6] the shield outer diameter fits comfortably inside the 94 mm PSXM envelope at both radii (table 1); the radial separation between the main-coil current locus and the ideal shield surface is 3.7 mm at the quadrupole threshold, a model separation that a real mechanical clearance would additionally include the conductor cross-section, insulation and supports. Table 4 collects the thresholds and the field delivered at each; the window is a starting geometry for the next stage of the study and not a verified design point, since the exterior-field side of the specification remains open.

Table 4: Shield-radius thresholds under the criteria considered. Each threshold is the smallest radius at which the stated target is deliverable at 1000 A in the present two-dimensional ideal-shell model. The BL -based dipole row is an illustrative sensitivity reading (effective length 0.1 m; section 6.4), not a design conclusion.

Mode	Criterion	Threshold R_s (mm)	Delivered at 1000 A
Quadrupole	1 mT/mm working benchmark	26.233	1.00 mT/mm
Dipole	1 mT central field (provisional)	22.990	1.00 mT
Dipole	BL at $L_{\text{eff}} = 0.1 \text{ m}$ (illustrative)	29.6	10.0 mT

The quadrupole result therefore supports $R_s = 26.2 \text{ mm}$ to 27.5 mm as the present baseline range, with the default geometry at the upper end; the dipole radius remains open (section 6.4). The quadrupole threshold bounds the radius from below, while the illustrative BL reading points to about 29.6 mm; the baseline range is therefore the smaller of the two only for as long as the dipole role and its effective length are unsettled, and it should be revisited before the geometry is frozen.

The scan varies the shield radius at fixed coil count and fixed coil geometry; the number of coils, the coil radius, the angular distribution and the shield wall thickness all remain at their baseline values. The coil count in particular is a stated design question left untouched here.

7 Further work

7.1 Physical shield response

The shield currents computed here are ideal equivalent surface currents. The physical chain is $I_{\text{main}}(t) \rightarrow B(t) \rightarrow -\partial_t B \rightarrow \mathbf{E} \rightarrow \mathbf{J}_{\text{shield}}$, so the shield current solves a circuit problem of the form

$$\mathbf{L}_{ss} \dot{\mathbf{I}}_s + \mathbf{R}_{ss} \mathbf{I}_s = -\mathbf{M}_{sm} \dot{\mathbf{I}}_m, \quad (18)$$

with resistance, self-inductance and mutual-inductance matrices $\mathbf{R}_{ss}$, $\mathbf{L}_{ss}$, $\mathbf{M}_{sm}$ (bold-face matrices are distinct from the scalar shield radius R_s and the physical length L used elsewhere). The solution carries the amplitude, phase lag and decay time of the physical response; the matrix S of eq. (6) contains only the spatial distribution and none of that. Solving eq. (18) and comparing the physical shielding factor with S is the calculation that would produce a shielding prediction, which this note does not contain.

The calculation requires hardware inputs that are not yet fixed. The PSXM working assumptions are a peak current of ~ 1 kA and a pulse width $\lesssim 1 \mu\text{s}$ [7]. Every eddy-current statement scales with the pulse frequency through the skin depth $\delta = \sqrt{2/\mu_0\sigma\omega}$, so the flux-exclusion approximation has to be assessed with the actual waveform; with the internal 5 mm copper baseline the skin depth is 0.06 mm at 1 MHz [6], consistent with the strong skin-effect regime, while the decay time and the Joule heat against the published $Q < 1$ W [1] still require the circuit model. The 1000 A cap used throughout is accordingly an adopted working-current cap rather than an established hardware limit.

7.2 Three-dimensional and remaining design work

The study is two-dimensional throughout, so a three-dimensional calculation is required to quantify the physical shielding factor and to settle the field-integral-based dipole radius of section 6.4; until it is available no exterior-field value quoted here should be read as performance. Interaction with the actual storage-magnet fringe field should likewise be evaluated once the field map at the PSXM location is available.

Only the shield radius has been varied. The number of coils, the coil radius and angular distribution, and the shield wall thickness remain at baseline; optimising the coil count would change the 30° ripple directly, since that ripple follows from the six-fold geometry, and is placed after the two items above because a geometry optimised against the present shield model would have to be redone once the induction model and the three-dimensional treatment change what the shield does. The outer magnet, and the extension to the actively shielded ASSM where the shield currents are controlled instead of induced so that S becomes a design choice, are outside the scope of this note; the question this project should ultimately answer is whether the passive alternative meets the same specifications as the published ASSM design [1] at lower complexity.

8 Conclusions

This note has studied the inner PSXM with a single method, an equal-weight least-squares inverse solve for the coil currents, in which the current locations and field-sampling locations are separate sets and the system is deliberately overdetermined. The implementation supports custom design weights, but none is used for the reported calculations. Within that model the six-coil ring has ample field headroom: at the 1000 A working-current cap the bare ring delivers 23.70 mT and 2.179 mT/mm, against the 1 mT and 1 mT/mm working benchmarks, and the useful field region is set by the coil radius rather than by the solver. The shield, and not the coil current cap, sets the operating point: at $R_s = 27.5$ mm the same ring delivers 7.836 mT and 1.203 mT/mm.

The quadrupole can be rolled continuously about the beam axis at very little cost. The currents at roll angle φ are $\cos 2\varphi$ and $\sin 2\varphi$ times two fixed patterns, verified against a full re-solve at every angle to solver round-off, so the magnet needs two stored patterns and one control parameter instead of a solve per angle. The amplitude ripples by 15.5 % with period 30° as a consequence of the six-fold coil geometry, while the field quality and the recovered roll angle are unaffected.

The shield radius is bounded from below by field delivery, and the bound has a closed form. Flux exclusion reduces the central amplitude of multipole order m by $1 - (R/R_s)^{2m}$, which inverts into an explicit design map for the radius (eq. (9)); the numerical scan reproduces it. The smallest radius at which the adopted 1 mT/mm quadrupole working benchmark is still deliverable at the 1000 A cap is $R_s \approx 26.2$ mm (26.233 mm in the numerical model). Both the benchmark and the current cap are adopted working assumptions rather than beam- or hardware-derived limits, so the threshold should be re-evaluated once they are fixed. For the dipole the threshold is not yet fixed at all: the provisional 1 mT central-field benchmark gives 22.990 mm, which is mechanically unbuildable and is best read as an artefact of that benchmark, while the published field integral $BL = 1.0 \times 10^{-3}$ T m cannot be converted into a central field without an effective magnetic length; read with the published 0.1 m length it implies about 10 mT and a radius of about 29.6 mm, an illustrative sensitivity estimate rather than a design conclusion. The dipole radius should be settled by evaluating the field integral in the finite-length geometry (section 6.4).

The shielding performance is the part this study does not deliver. The continuum limit of the model is an exactly flux-excluding shell whose exterior field vanishes for zero net enclosed current, so every exterior-field value computed here is a discretisation residual relative to that continuum solution, and the published $1 \mu\text{T}$ requirement can therefore not be verified here. The mechanisms that will decide it are all outside the model: imperfect flux exclusion at finite conductivity and pulse length, the open ends of a can of finite length, and any longitudinal slit. Closing that gap means, in order, confirming the real pulse and material parameters, building the induction model of eq. (18), and evaluating the shielding factor in three dimensions (section 7). What this note establishes is the field capability of the six-coil geometry, the cost of rolling the quadrupole, and the closed-form shield counter-field penalty that sets a lower bound on the shield radius.

A Reproducing the results

The derived figures and numerical results in this note come from the `PSXM_sim` codebase [8]. The analysis scripts write the key reported outputs as LaTeX macros that this document inputs, while fixed modelling assumptions and published specifications remain explicit in the source; `report_scan.py` performs the shield-radius scan once and caches it, and `report_figures.py` draws every figure from that cache and from the same solver modules, so figures and tables cannot disagree.

Result	Script
Solver and analytic cross-check	<code>current_solver.py</code> , <code>verify_ls.py</code>
Field capability	<code>field_capability.py</code> , <code>field_analysis.py</code>
Rotational quadrupole	<code>rot_quad.py</code>
Ideal shield and convergence	<code>shield.py</code> , <code>report_robust.py</code>
Shield-radius scan	<code>report_scan.py</code>
Field quality	<code>report_background.py</code>
Report figures	<code>report_figures.py</code>

Figure 1 comes from `tracker/analyze_track.py`, which requires the tracking data `trk20000.root`, a large binary file that is not part of the repository.

B Ideal-shell multipole result

Throughout, $\mathbf{B}$ denotes the incremental pulsed field of the PSXM, defined relative to the static storage-solenoid background; the static background is not part of the inverse problem and is unaffected by the shield currents. For an infinitely long cylindrical shell in the perfect-flux-exclusion limit, the incremental flux cannot penetrate the wall, so the boundary condition on its surface is

$$\Delta B_n = \Delta B_x \cos \theta + \Delta B_y \sin \theta = 0, \quad (19)$$

which for a two-dimensional field is the same as $A_z = \text{const}$ on the shell. Matching the interior source's cylindrical multipoles against the surface-current sheet order by order, a standard result of the cylindrical-multipole expansion [9], shows that for zero net enclosed current every exterior multipole coefficient is cancelled by the induced sheet, so the exterior incremental field vanishes identically. Requiring $\Delta \mathbf{B} = 0$ outside the shield and requiring eq. (19) on it therefore describe the same ideal shield; the free-space rings are the better-conditioned way to impose it with a discrete current sheet (section 5). This is the origin of the central-amplitude reduction factor of eq. (7).

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